Reconstitution of DANTRIUM® Intravenous (dantrolene sodium)

Important, read before use

The filter needles provided with this product are intended for use when drawing reconstituted Dantrolene Sodium IV solution from the vial into the syringe prior to administration to the patient. The filter needle then needs to be **exchanged** for an appropriate intravenous cannula or giving set and MUST NOT be used to administer the product to the patient.

Filter needles are only provided with newly issued stock, existing stock is not affected and does not require filtering.

NOT FOR SKIN INJECTION 18G X 1 1/2

Filter Needles Provided

This box contains 12 BD Blunt Fill Needles with Filter

18G x 1 ½ (1.2mm x 40mm) (5µm)

(BD product code 305211)

Not for skin injection

Do not autoclave BD Blunt Filter (Fill) Needle

Sterile and non-pyrogenic. Do not use if individual packaging is damaged

Caution: Re-use may lead to infection or other illness/injury

Discard entire product in approved sharps collector

Instructions for use of the Blunt Fill Needles with Filter

(to be performed in accordance with local infection control guidelines):

1) Reconstitute the vial with 60 mL of Water for Injection.
2) Filter the reconstituted product with the blunt fill needle with filter provided when drawing up the solution into the syringe.
3) Remove the blunt fill needle from the syringe prior to attachment to an intravenous cannula or giving set.
4) Discard the blunt fill needle and product vial in an approved sharps collector.
5) Use a new filter needle with every vial of DANTRIUM® I.V.
6) Administer DANTRIUM® I.V. immediately upon reconstitution.

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PHYSICIAN LEAFLET

DANTRIUM® Intravenous 20 mg Powder for Solution for Injection

(dantrolene sodium)

Indications
For the treatment of malignant hyperthermia.

Mode of Action
In isolated muscle preparations, dantrolene sodium uncouples the excitation and contraction of skeletal muscle, probably by interfering with the release of calcium from sarcoplasmic reticulum. In the malignant hyperthermia syndrome, evidence points to an intrinsic abnormality of muscle tissue. It has been postulated that "triggering agents" (i.e. skeletal muscle relaxants and inhalation anaesthetics) induce a sudden rise in myoplasmic calcium either by preventing the sarcoplasmic reticulum from accumulating calcium adequately, or by accelerating its release. This rise in myoplasmic calcium activates the acute catabolic processes involved in the malignant hyperthermia crisis. Dantrolene sodium may prevent the increases in myoplasmic calcium and the acute catabolism within the muscle cell by interfering with the release of calcium from the sarcoplasmic reticulum to the myoplasm. Thus the physiological, metabolic and biochemical changes associated with the crisis may be reversed or attenuated.

Dosage and Administration
As soon as the malignant hyperthermia syndrome is recognised all anaesthetic agents should be discontinued. An initial Dantrium IV dose of 1 mg/kg should be given rapidly directly into the vein.

It must not be mixed with other intravenous infusions. If the physiological and metabolic abnormalities persist or reappear, this dose may be repeated up to a cumulative dose of 10 mg/kg. Clinical experience to date has shown that the average dose of Dantrium IV required to reverse the manifestations of malignant hyperthermia has been 2.5 mg/kg. If a relapse or recurrence occurs, Dantrium IV should be readministered at the last effective dose.

For intravenous use only.

Mixing Instructions
Each vial of Dantrium IV should be reconstituted by adding 60 ml Water for Injections and shaking until the solution is clear.

Precautions and Warnings

a. In some subjects as much as 10 mg/kg of Dantrium IV has been needed to reverse the crisis. In a 70 kg man this dose would require approximately 36 vials. Such a volume has been administered in approximately one and a half hours.
b. Contraindication: Hypersensitivity to dantrolene sodium or to any of the excipients.
c. When mannitol is used for prevention or treatment of renal complication of malignant hyperthermia the 3000 mg of mannitol needed to dissolve each vial of intravenous dantrolene sodium (20mg) should be taken into consideration.
d. Because of the high pH of the intravenous formulation of Dantrium and potential for tissue necrosis, care must be taken to prevent extravasation of the intravenous solution into the surrounding tissues.
e. Pregnancy: the safety of Dantrium Intravenousin pregnant women has not been established; Dantrolene sodium crosses the placenta and it should be given only when the potential benefits have been weighed against the possible risk to mother and child.
f. Dantrolene has been detected in human milk at low concentrations (less than 2 micrograms per millilitre) during repeat intravenous administration over 3 days. Dantrium Intravenous should be used by nursing mothers only if the potential benefit justifies the potential risk to the infant.
g. The use of Dantrium Intravenous in the management of malignant hyperthermia is not a substitute for previously known supportive measures. It will be necessary to discontinue the suspected triggering agents, attend to increased oxygen requirements and manage the metabolic acidosis. When necessary institute cooling, attend to urinary output and monitor for electrolyte imbalance.
h Hepatic dysfunction, including hepatitis and fatal hepatic failure, has been reported with dantrolene sodium therapy. Whilst the licensed indications of intravenous dantrolene sodium do not generally necessitate prolonged therapy, the risk of hepatic dysfunction may increase with dose and duration of treatment, based on experience with oral therapy. However in some patients it is of an idiosyncratic or hypersensitivity type, and could occur after a single dose.
i. A decrease in grip strength and weakness of leg muscles, especially walking down stairs,can be expected post-operatively. In addition, symptoms such as "lightheadedness" may be noted. Since some of these symptoms may persist for up to 48 hours, the patient must not operate an automobile or engage in other hazardous activity during this time.

Interactions with Other Drugs
The combination of therapeutic doses of intravenous dantrolene sodium and verapamil in halothane/-alphachloralose anaesthetised swine has resulted in ventricular fibrillation and cardiovascular collapse in association with marked hyperkalaemia. Hyperkalaemia and myocardial depression have also been reported rarely in malignant hyperthermia-susceptible patients receiving intravenous dantrolene and concomitant calcium channel blockers. Hence, the use of Dantrium IV and calcium channel blockers in combination is not recommended. Administration of dantrolene sodium may potentiate vecuronium-induced neuromuscular block.

Side Effects and Adverse Reactions
There have been occasional reports of death following malignant hyperthermia crisis even when treated with intravenous dantrolene sodium; incidence figures are not available (the pre-dantrolene sodium mortality of malignant hyperthermia crisis was approximately 50 %). Most of these deaths can be accounted for by late recognition, delayed treatment, inadequate dosage, lack of supportive therapy, intercurrent disease and/or the development of delayed complications such as renal failure or disseminated intravascular coagulopathy. In some cases there are insufficient data to completely rule out therapeutic failure of dantrolene sodium.

The administration of intravenous dantrolene sodium to human volunteers is associated with loss of grip strength and weakness in the legs, as well as subjective central nervous system complaints.

There are rare reports of pulmonary oedema developing during the treatment of malignant hyperthermia crisis in which the diluent volume and mannitol needed to deliver intravenous dantrolene sodium possibly contributed.

Extravasation may lead to tissue necrosis.

Hepatic dysfunction may occur, including fatal hepatic failure (see section "Precautions and warnings").
Tabulated list of adverse other reactions
Frequency cannot be estimated from available data.

System Organ Class	Frequency	Adverse drug Reactions
Nervous system disorders	Unknown	Dizziness, somnolence, convulsion speech disorder
Cardiac disorders	Unknown	Cardiac failure, bradycardia, tachycardia
Respiratory, thoracic and mediastinal disorders	Unknown	Pulmonary oedema (dilutent volume and mannitol needed to deliver dantrolene IV may contribute to the event), Pleural effusion, respiratory failure, respiratory depression
Gastrointestinal disorders	Unknown	Abdominal pain, nausea, vomiting, gastrointestinal bleeding
Hepatobiliary disorders	Unknown	Hepatic dysfunction including fatal hepatic failure (see section 4.4), jaundice, hepatitis
Skin and subcutaneous disorders	Unknown	Hyperhidrosis
Renal and urinary disorders	Unknown	Crystalluria
General disorders and administration site conditions	Unknown	Rash, erythema, localised pain and thrombophlebitis

Shelf-Life
Three years. The reconstituted solution should be used within six hours.

Excipients
Mannitol
Sodium hydroxide

Storage Precautions
Unopened product: Do not store above 25°C. Reconstituted product: Store between 15 and 25°C.
Do not refrigerate or freeze. Protect from direct light.

Marketing Authorisation Holder and Manufacturer:
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Legal Category: POM

This leaflet was last revised in 04/2016